AEROO SPACE AI COMPETITION

Planet Finder

A neural network for exoplanet detection from stellar light curves

Dossymbekov Aslan, 9th grade
Specialized Lyceum 165, Almaty
«AslanD» Team

5 000+
stars were used to train the AI

< 10 ms
prediction time

~90%
model accuracy

The universe is vast. There is too much data.

Humanity is accumulating terabytes of astronomical data faster than it can analyze it.

200 000+

Stars in the Kepler database
Manual processing would take hundreds of years

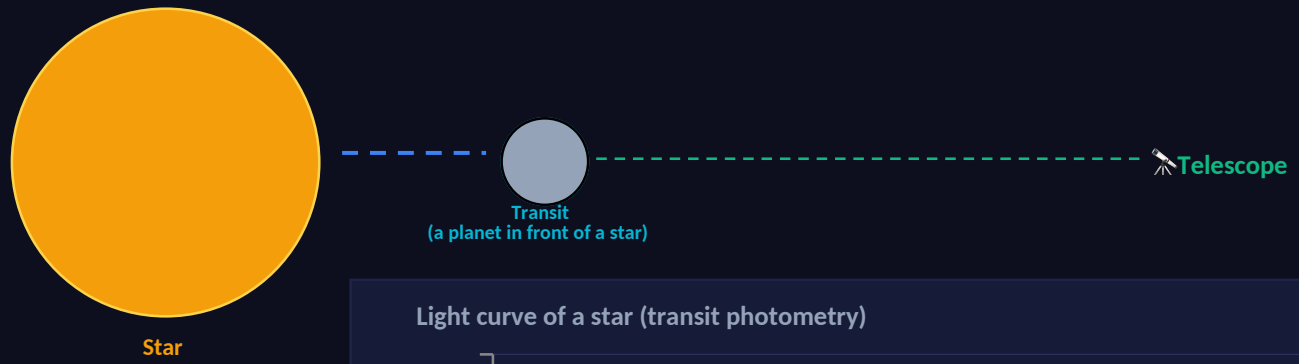
~3000

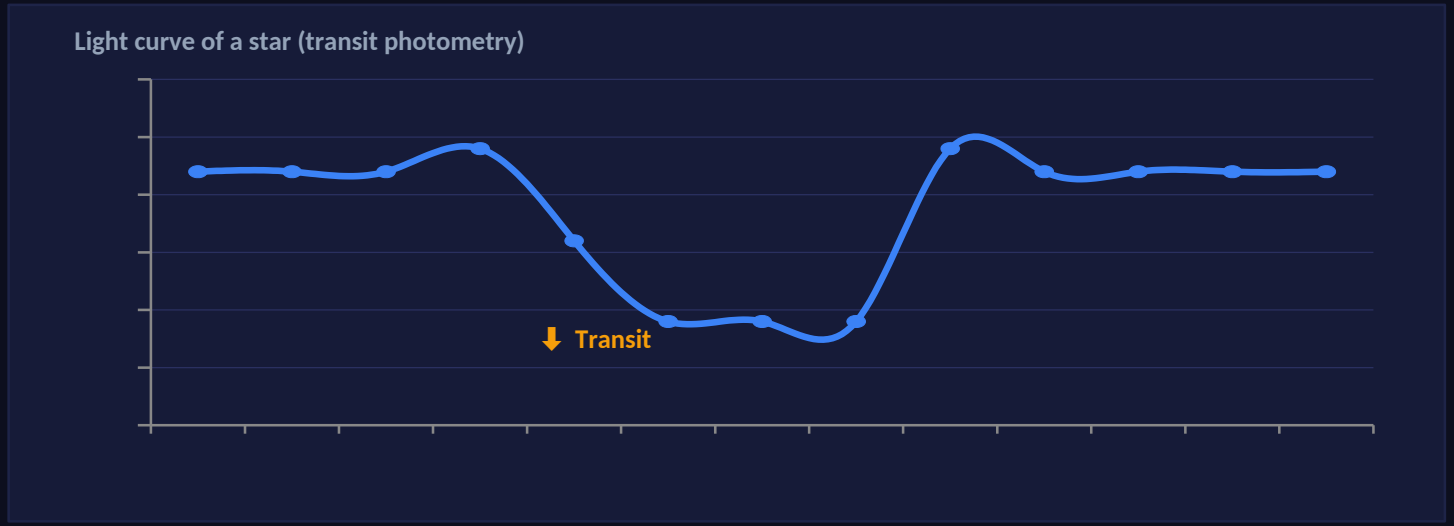
planets confirmed
astronomers physically don't have time to view them

20%

of transit signals are missed
Due to human factor

How does the transit method of planet detection work?





Transit photometry method + LSTM

1

downloading NASA data
Kepler/TESS

2

Preprocessing
Removing NaN, normalization

3

Training a neural network

4

Ready to use!

Input
[brightness, time]
→
1 LSTM layer
64 hidden units
→
Linear
Input: 64
Output: 0-9 planets

Artificial intelligence at the core of the product

LSTM — Long Short-Term Memory

A recurrent neural network specifically designed for time series analysis.

- Remembers transit patterns spaced over weeks and months
- Handles variable-length sequences
- Robust to noise in real astrophysical data
- Outperforms simple RNNs on long-term dependencies

- **Rust — perfomance and memory safety**
- **LibTorch - PyTorch C++ API**
- **NASA/TESS/Kepler from lightkurve pip**
- **CUDA (optional) — faster by 10 times**
- **Apache 2.0 license**

The working prototype is now complete.

PlanetFinder — Terminal

```
$ cargo run --release predict
   Compiling planetfinder v1.0.1
   Finished [optimized] target(s)



   Loaded model: model.ot
>> Enter 'brightness time'. end = result
   0.998  131.2
   0.872  134.8  ← transit!
   0.999  136.6
   end

>> Predicted number of planets: 1
```

MVP Status

- ✓ Trained on NASA data
- ✓ Prediction < 10 ms
- ✓ CUDA GPU-acceleration
- ✓ Windows / Linux / Mac
- ✓ Github repositoty
- ✓ Working website

https://planetfinder.online — live web demo

Server

Debian 13 · Frankfurt, Germany
EU infrastructure

Perfomance

Prediction < 30 ms

Security

Rate limiting: 5 requests/minute for 1 IP

Already done: working MVP on GitHub

- ✓ Full featured CLI tool
- ✓ Open Source
- ✓ NASA data (Kepler, TESS)
- ✓ Ready-to-use trained model
- ✓ Detailed documentation
- ✓ Веб-демо

Competitor analysis

Solution	Language	Open Source	LSTM	Perfomance	NASA data
PlanetFinder	Rust	✓	✓	< 10 ms	✓
AstroNet (Google)	Python	✓	✗	~200 ms	✓
ExoNet (MIT)	Python	✗	✗	~500 ms	✗
Lightkurve ML	Python	✓	✗	~150 ms	✓
Manual analysis	—	—	—	few days	✓

AI vs reality

Kepler-11

Distance: ~2 000 light years from the Earth

Reality (NASA)

6

Confirmed planets

=

PlanetFinder

6

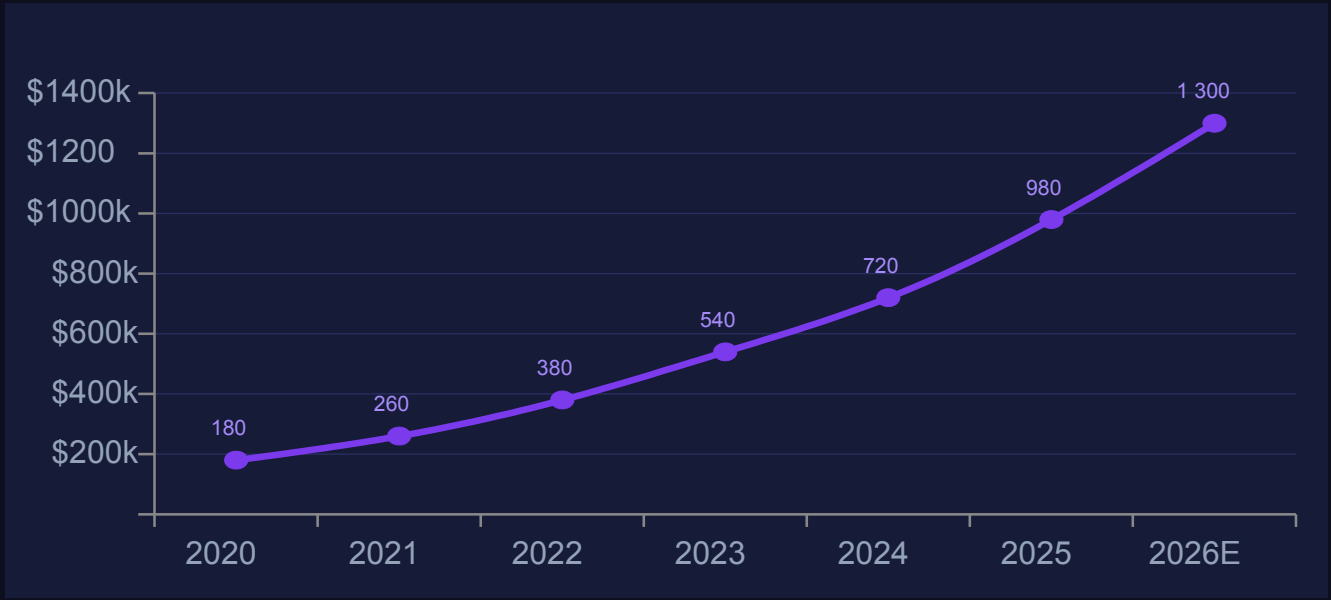
The neural network prediction

```
$ planet_finder predict
>> flux time (end for result)
0.9981  134.2
0.9721  138.7  ← transit b
0.9854  141.3
0.9612  145.9  ← transit c
0.9901  150.1
0.9743  153.4  ← transit d
... (500 data points)
end
>> Predicted number of planets: 6
```

Planets of Kepler-11:

- Kepler-11b
- Kepler-11c
- Kepler-11d
- Kepler-11e
- Kepler-11f
- Kepler-11g

The SpaceTech AI market is growing by 18% annually.



150+
active space missions

$2.5B
Investments in space by 2025

Sources: NASA, ESA, Space Foundation Report 2025, Bloomberg Intelligence

Monetization and ways to market

Open Source Core

Free

- GitHub Community
- Basic CLI
- Training on TXT
- Apache 2.0

Pro API

$50 / month

- REST API acces
- Batch-processing
- Cloud GPU
- SLA 100%

Enterprise

From $1 000

- Custom models
- On-premise
- Integration with software
- 24/7 support

Grants

Free

- Startups
- Universities
- Observatories
- Grants

Development strategy 2026–2028

1

Q1–Q2 2026

MVP

- ✓ LSTM model on Rust)
- ✓ NASA data
- ✓ Open Source
- ✓ Documentation & GitHub
- ✓ Web demo

2

Q3–Q4 2026

Product

- REST API (cloud)
- Accounts system
- Batch processing
- First 50 clients
- Payment system

3

2027

Growth

- Enterprise
- On-premise
- Live integration with ESA/NASA
- 10+ observatories
- Seed round

4

2028

Scale

- Live integration with TESS/JWST
- Glocal partners
- Team
- Series A
- Stable income

5 reasons to invest in PlanetFinder

1

Unique technology

The only open source exoplanet detector on Rust and LSTM

2

Huge market

SpaceTech AI: $4.2B TAM

3

Working MVP

Not a concept
But a living product

4

Scientific basis

Real NASA, Kepler and TESS data

5

Perfect timing

JWST is running. TESS is active.
Billions of new data points.

Let's find planets together.

Invest in technology that literally expands the boundaries of the known universe.

Thank you for your attention!